

Student-based Evaluation of LLM-generated Ontology Design Patterns

Supplementary material to the ODPGen study

Overview

This supplementary report summarises the student evaluation conducted alongside the main expert evaluation of LLM-generated Ontology Design Patterns (ODPs). Students were recruited from two Knowledge Engineering courses and completed a dedicated form, distinct from the expert form, which focused on aspects more relevant to less experienced users (clarity of documentation and of naming, time required to understand a draft).

A total of 28 responses were collected from 7 students. Each student was randomly assigned two of the patterns in the dataset and rated both the Creativity-track output (*Llama-3.1-8B, scenario-only*) and the Functionality-track output (*Llama-2-70B, cq-only*) for every assigned pattern, yielding four assessments per student. The pool covers 9 of the 14 patterns in the dataset.

Question schema

Each response consisted of the following items:

- **Student level.** Self-reported background (MSc or PhD).
- **Time to understand.** Free-text answer, in minutes, for the time required to understand the generated draft.
- **Documentation clarity (1–5).** Likert rating of how clear the natural-language documentation is.
- **Naming clarity (1–5).** Likert rating of how clear and consistent class and property names are.
- **Adoption threshold.** Whether the student would start from the generated draft (*Yes / Maybe / No*), with a brief justification.
- **Time saved vs. from-scratch.** Estimated proportion of time saved if the draft were used as a starting point (0%, 1–25%, 26–50%, 51–75%, 76–100%).

Results by track

Across both tracks, students rate the documentation clarity at 3.14/5 on average and the naming clarity at 3.96/5. Adoption answers are skewed towards *Maybe*, with comparatively few outright *Yes* answers.

Table 1: Student evaluation by track. Mean documentation clarity and mean naming clarity are reported on a 1–5 Likert scale; adoption percentages and mean estimated time saved (interval midpoints) are derived from the same forms.

Track	<i>N</i>	Doc. clarity	Naming clarity	Yes%	Maybe%	Time saved
Creativity	14	3.14	4.14	29%	50%	30.7%
Functionality	14	3.14	3.79	7%	50%	21.9%
Overall	28	3.14	3.96	18%	50%	26.3%

Table 2: Student evaluation by self-reported level (MSc / PhD). The mean time-to-understand is in minutes and is derived from the leading numeric value of each free-text answer.

Level	<i>N</i>	Doc. clarity	Naming clarity	Yes%	Time-to-understand (min)
MSc	12	3.42	3.83	25%	4.8
PhD	16	2.94	4.06	12%	5.4

Results by self-reported level

Per-pattern results

Note on the data

Students provided written justifications for their adoption choices. Recurring themes (missing properties, weak documentation, overly literal reuse of scenario vocabulary) align with those identified in the main expert pool and are not reproduced here. Free-text responses are not redistributed alongside this summary in order to protect respondent anonymity.

Table 3: Student evaluation by pattern and track. Doc. clarity and Naming clarity are 1–5 Likert means; Yes% is the share of *Yes* adoption answers.

Pattern	Creativity (<i>scenario-only</i>)				Functionality (<i>cq-only</i>)			
	<i>N</i>	Doc.	Naming	Yes%	<i>N</i>	Doc.	Naming	Yes%
Causality (US 1)	1	3.00	4.00	0%	1	3.00	4.00	0%
Solid Apps Governance: Data Request	3	4.00	4.33	67%	3	3.67	3.67	0%
Solid Apps Governance: Data Log	1	2.00	4.00	0%	1	3.00	3.00	0%
Role-Dependent Names	3	3.00	5.00	33%	3	2.00	3.33	0%
IndiLegalOnt	1	1.00	2.00	0%	1	3.00	4.00	0%
AISHIP	1	5.00	5.00	0%	1	5.00	4.00	0%
ARGOS	1	4.00	4.00	0%	1	4.00	4.00	100%
MQTT4SSN	1	2.00	5.00	100%	1	2.00	5.00	0%
ODRL Access Control: State of the World	2	3.00	3.00	0%	2	3.50	4.00	0%